FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

Chapter 6: Three-Phase AC Circuits

Three-Phase AC Circuits Comprehensive Final Exam Mastery Guide

COURSE & DEPT	INSTRUCTOR	COVERAGE
Fundamental Electrical Engineering (EE KKU)	Assoc. Prof. Dr. Nararat Ruangchaijatupon	16 Diagrams • 5 Exam Problems • Full Mathematical Proofs

Chapter 6: Three-Phase Alternating Current Circuits (Comprehensive Mastery Guide)

Course: Fundamental Electrical Engineering (FunE) • Faculty of Engineering, Khon Kaen University
Instructor: Assoc. Prof. Dr. Nararat Ruangchaijatupon

📌 Introduction: Why Three-Phase AC Power? (Intuition from First Principles)

Before memorizing formulas and subscripts, let us understand the physical and economic intuition:

1. **Single-Phase Power (1ϕ)**: The standard household supply ($230\ \text{V}$) is like **"pedaling a bicycle with only one leg"**. When the crank reaches the top and bottom dead centers, instantaneous power drops to zero ($p(t)$ pulsates at twice the supply frequency, 2ω). Consequently, large single-phase motors vibrate, run unevenly, and lack inherent starting torque.
2. **Three-Phase Power (3ϕ)**: Three identical coils positioned 120° apart in space inside an electrical alternator. It acts like a **"three-cylinder engine firing sequentially in perfect balance"**:
   - **Constant Instantaneous Power ($p(t) = \text{Constant}$)**: In a balanced three-phase system, the total instantaneous power delivered to the load never drops to zero. Large motors operate smoothly without mechanical vibrations and produce self-starting rotating magnetic fields.
   - **Massive Conductor Material Savings**: Delivering a given amount of power over a transmission line requires $25\% - 50\%$ **less copper or aluminum** in a three-phase system compared to a single-phase system carrying the same power at equivalent losses.
   - **Two Distinct Voltage Levels from a Single System**: A standard 3-phase 4-wire system provides both $230\ \text{V}$ (Phase Voltage, line-to-neutral for lights and appliances) and $400\ \text{V}$ (Line Voltage, line-to-line for heavy machinery and industrial motors).

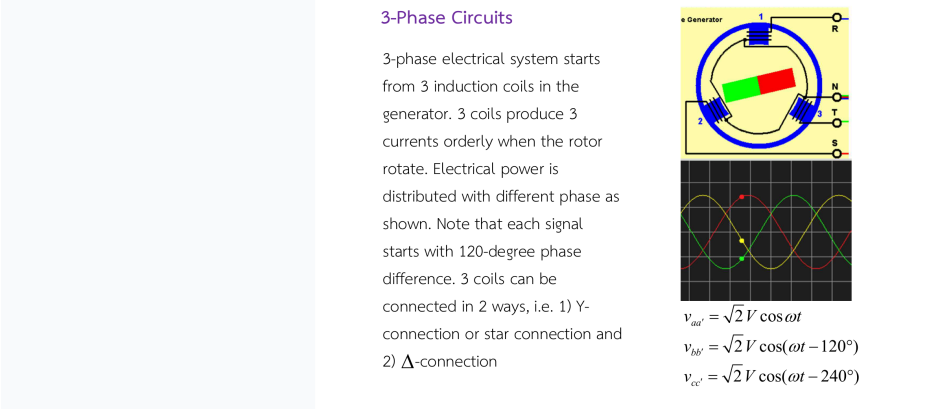


Figure 1: Generation of Three-Phase Voltages and Waveforms

Figure 1: Generation of 3-phase voltages from three coils displaced by $120°$ in the generator stator, producing three equal sinusoidal waveforms phase-shifted by $120°$ (From Slide 3).

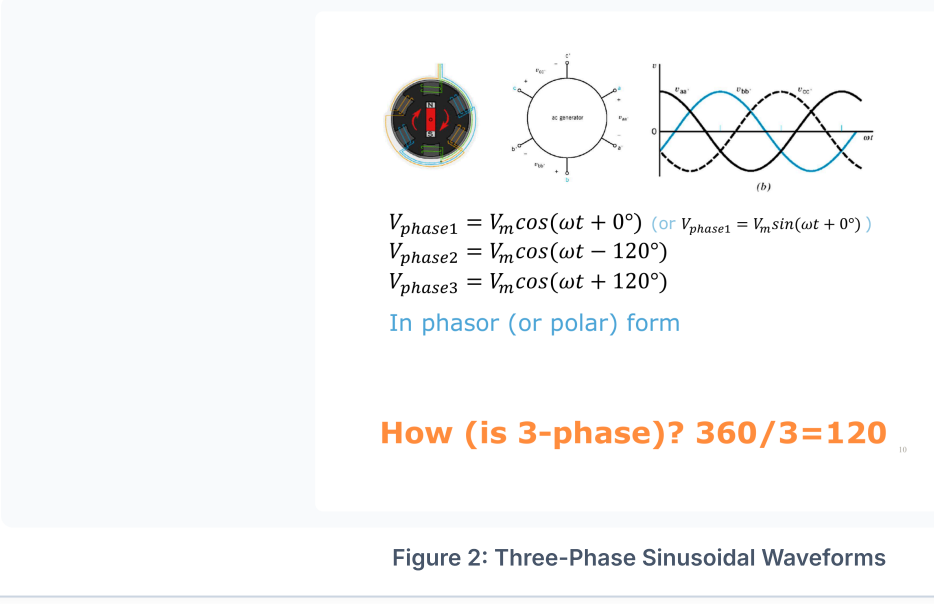
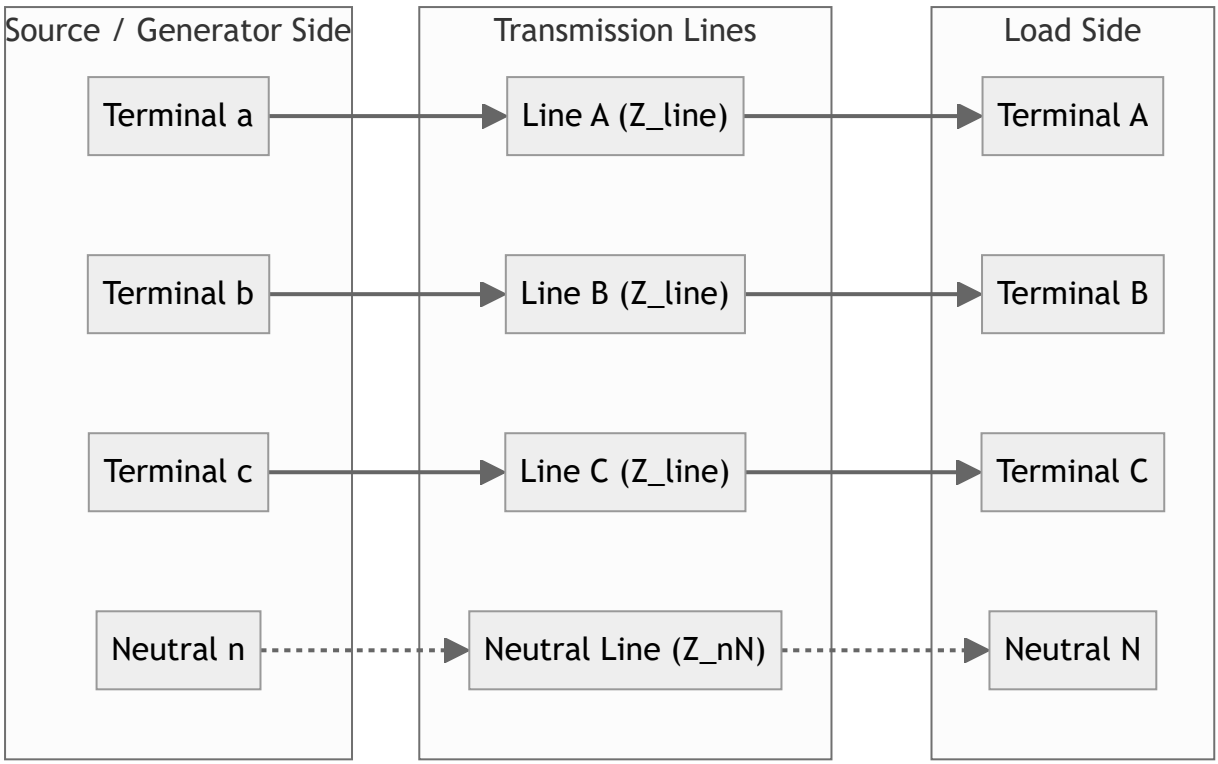


Figure 2: Three-Phase Sinusoidal Waveforms

Figure 2: The three sinusoidal waveforms (v_a, v_b, v_c) with period $T = 360°$, each shifted by $360°/3 = 120°$ (From Slide 10).

🔤 Complete Subscript & Notation Decoder (Variables Decoded)

The primary reason students struggle with three-phase circuits is confusing subscripts. Let us clarify them permanently:



1. Meaning of Subscript Letters

- **Lowercase Letters (a, b, c, n)**: Represent the **Source / Generator side**, where n is the source neutral point.
- **Uppercase Letters (A, B, C, N)**: Represent the **Load / Receiver side**, where N is the load neutral point.
- **Double Subscripts Represent Potential Difference**:
  - $\mathbf{V}_{an} = \mathbf{V}_a - \mathbf{V}_n$: Voltage from source terminal a to source neutral n (**Source Phase Voltage**).
  - $\mathbf{V}_{AN} = \mathbf{V}_A - \mathbf{V}_N$: Voltage across load phase A to load neutral N (**Load Phase Voltage**).
  - $\mathbf{V}_{ab} = \mathbf{V}_a - \mathbf{V}_b$: Voltage between transmission line a and line b (**Line-to-Line Voltage** or **Line Voltage**).
  - $\mathbf{V}_{Nn} = \mathbf{V}_N - \mathbf{V}_n$: Voltage difference between load neutral and source neutral (**Neutral Shift Voltage**).

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2. Line vs. Phase Quantities Comparison Table

Quantity	Symbol	Physical Definition	In Star / Wye (Y) Circuit	In Delta (Δ) Circuit
Phase Voltage	V_p or V_{phase}	Voltage dropped across **one single coil/branch**	Measured from phase terminal to neutral ($\mathbf{V}_{an}$)	Measured across individual delta branch ($\mathbf{V}_{ab}$)
Line Voltage	V_L or V_{line}	Voltage measured between **any two transmission lines**	Measured line-to-line, e.g. $\mathbf{V}_{ab}, \mathbf{V}_{bc}, \mathbf{V}_{ca}$	Measured line-to-line, e.g. $\mathbf{V}_{ab}, \mathbf{V}_{bc}, \mathbf{V}_{ca}$
Phase Current	I_p or I_{phase}	Current flowing through **the internal coil winding**	Current through single phase arm ($I_p = I_L$)	Current circulating in delta branch ($\mathbf{I}_{ab}$)
Line Current	I_L or I_{line}	Current travelling along **the external transmission line wire**	Current in transmission line ($\mathbf{I}_{aA}, \mathbf{I}_{bB}, \mathbf{I}_{cC}$)	Current in transmission line ($\mathbf{I}_a, \mathbf{I}_b, \mathbf{I}_c$)

🧭 Phase Sequence: Positive (abc) vs. Negative (acb)

In a balanced system, all three phase voltages have equal magnitude (V_p) and are displaced by 120°:

1. Positive Sequence (abc Sequence) — *Standard in 95% of Problems*

The phasor rotates counter-clockwise. Phase a passes the reference axis first, followed by phase b, then phase c:

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$

$$\mathbf{V}_{bn} = V_p \angle -120^\circ$$

$$\mathbf{V}_{cn} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

2. Negative Sequence (acb Sequence) — *Used in Exam Trap Questions*

Phase a passes first, followed by phase c, then phase b:

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$

$$\mathbf{V}_{bn} = V_p \angle +120^\circ$$

$$\mathbf{V}_{cn} = V_p \angle -120^\circ$$

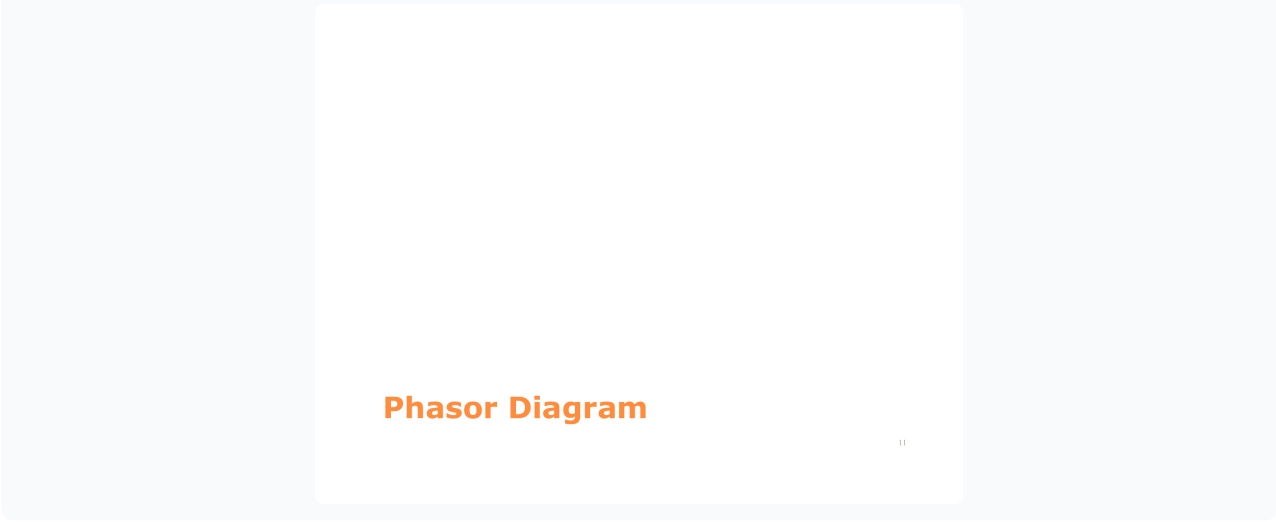
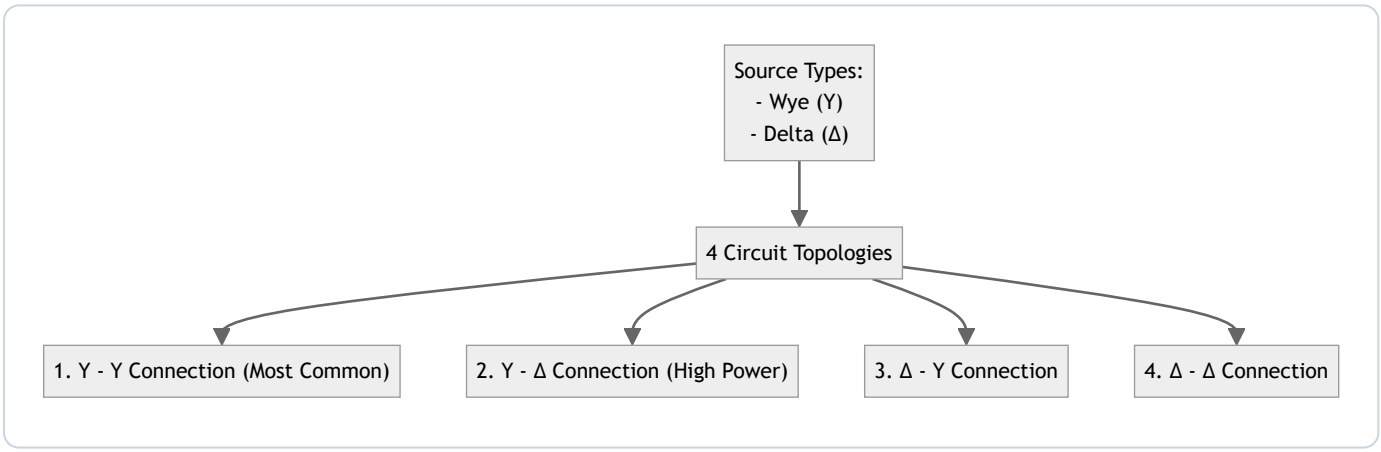


Figure 3: Phasor Diagrams for Balanced Three-Phase Voltages

Figure 3: Phasor diagrams of positive (abc) vs negative (acb) phase sequences (From Slide 11).

🔗 The 4 Interconnection Topologies (Source-to-Load)

Both source and load can be connected in either Star (Y) or Delta (Δ). This yields four possible topologies:



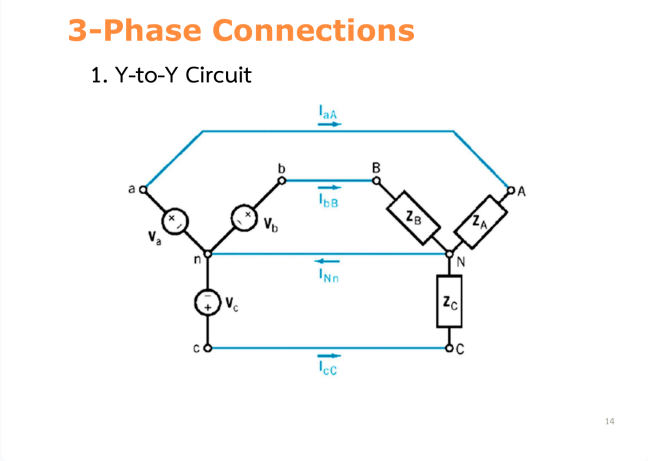


Figure 4: Wye-to-Wye (Y-Y) Connection System

Figure 4: A 3-phase 4-wire Y-Y connected system showing source, line impedance, and star-connected load (From Slide 14).

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1. Mathematical Proof of Wye (Y) System: $V_L = \sqrt{3}V_p\angle +30^\circ$

In a Wye-connected source, applying Kirchhoff's Voltage Law (KVL) between terminal a and terminal b:

$$\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn}$$

For positive sequence with $\mathbf{V}_{an} = V_p\angle 0^\circ$ and $\mathbf{V}_{bn} = V_p\angle -120^\circ$:

$$\mathbf{V}_{ab} = V_p(1+j0) - V_p\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = V_p\left(\frac{3}{2} + j\frac{\sqrt{3}}{2}\right)$$

Converting to polar form:

$$|\mathbf{V}_{ab}| = V_p\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = V_p\sqrt{\frac{9+3}{4}} = V_p\sqrt{3} = \sqrt{3}V_p$$

$$\theta = \arctan\left(\frac{\sqrt{3}/2}{3/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$$

$$\mathbf{V}_{ab} = \sqrt{3}V_p\angle +30^\circ$$

$$\mathbf{V}_{bc} = \sqrt{3}V_p\angle -90^\circ$$

$$\mathbf{V}_{ca} = \sqrt{3}V_p \angle +150^\circ$$

IMPORTANT

Summary for Wye (Y) Connection:

- **Line Voltage leads Phase Voltage by 30°:** $V_L = \sqrt{3}V_p \angle +30^\circ$
- **Line Current equals Phase Current:** $I_L = I_p$

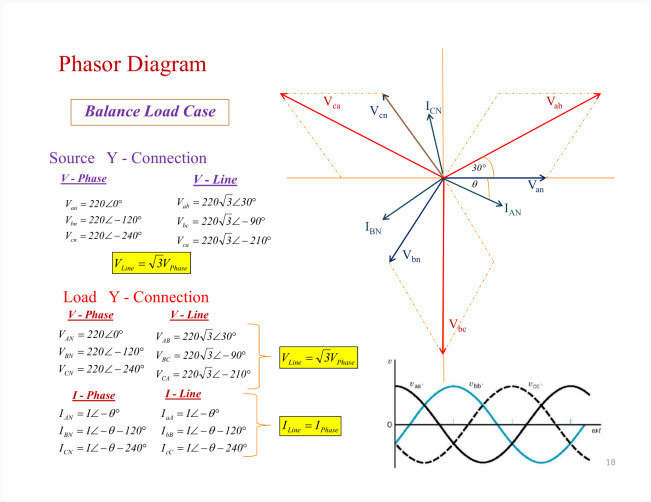


Figure 5: Phasor Proof that Line Voltage Leads Phase Voltage by 30 Degrees

Figure 5: Graphical phasor proof showing $\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn} = \sqrt{3}V_p \angle 30^\circ$ (From Slide 18).

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2. Mathematical Proof of Delta (Δ) System: $I_L = \sqrt{3}I_p \angle -30^\circ$

In a Delta connection, the phase branches are connected directly between the line conductors:

$$V_L = V_p$$

Applying Kirchhoff's Current Law (KCL) at terminal node A:

$$\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA}$$

Assuming balanced positive sequence currents $\mathbf{I}_{AB} = I_p \angle 0^\circ$ and $\mathbf{I}_{CA} = I_p \angle +120^\circ$:

$$\mathbf{I}_a = I_p(1 + j0) - I_p\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = I_p\left(\frac{3}{2} - j\frac{\sqrt{3}}{2}\right) = \sqrt{3}I_p \angle -30^\circ$$

IMPORTANT

Summary for Delta (Δ) Connection:

- **Line Voltage equals Phase Voltage:** $V_L = V_p$
- **Line Current lags Phase Current by** $30°$**:** $I_L = \sqrt{3} I_p \angle -30°$

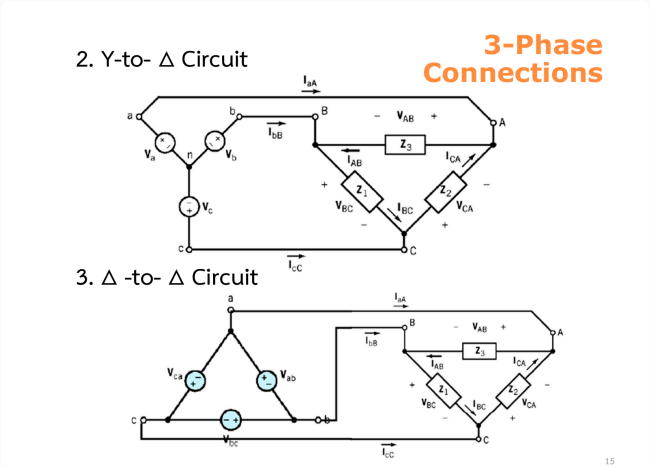


Figure 6: Wye-to-Delta (Y-Δ) Connection System

Figure 6: Wye source supplying a Delta-connected load (From Slide 15).

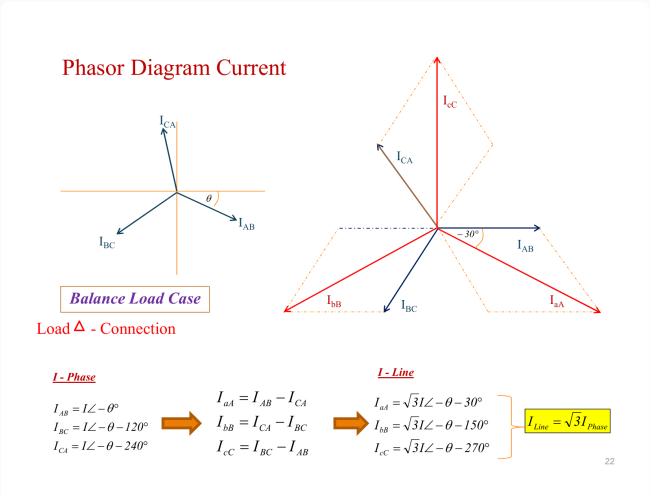


Figure 7: Phasor Proof for Delta Line and Phase Currents

Figure 7: Phasor relationship showing $\mathbf{I}_a = \sqrt{3} I_{AB} \angle -30°$ *(From Slide 22).*

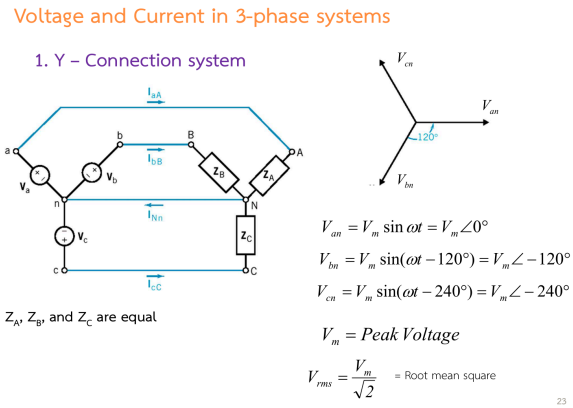


Figure 8: Master Summary of Wye and Delta Relationships

Figure 8: Comprehensive summary table of Line vs Phase quantities for Wye and Delta connections (From Slide 23).

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3. Delta-Wye Impedance Transformation ($\Delta \leftrightarrow Y$)

When solving balanced circuits containing Delta loads, transform the Delta load into an equivalent Wye load to solve on a **Per-Phase Equivalent** basis:

$$Z_Y = \frac{Z_\Delta}{3} \iff Z_\Delta = 3Z_Y$$

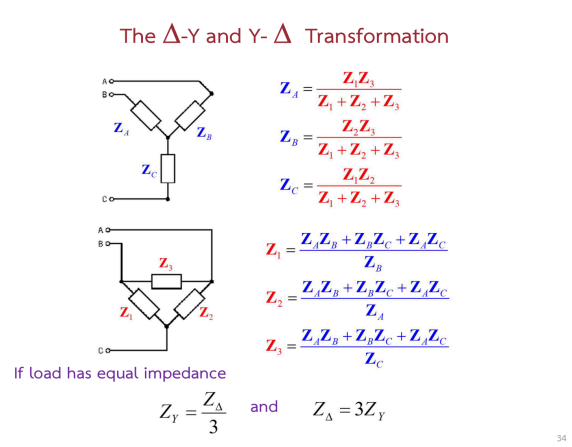


Figure 9: Delta to Wye Impedance Transformation

Figure 9: Transforming a balanced Delta load to an equivalent Wye load $Z_Y = Z_\Delta/3$ (From Slide 34).

Power in Three-Phase Circuits

The total power consumed by a balanced three-phase load (whether connected in Wye or Delta) is three times the single-phase power:

1. Complex, Real, and Reactive Power Formulas

$$\mathbf{S}_{3\phi} = 3\mathbf{S}_{1\phi} = 3\mathbf{V}_p\mathbf{I}_p^* = \sqrt{3}V_L I_L \angle\theta$$

$$P_{3\phi} = 3V_p I_p \cos\theta = \sqrt{3}V_L I_L \cos\theta \quad [\text{Watts, W}]$$

$$Q_{3\phi} = 3V_p I_p \sin\theta = \sqrt{3}V_L I_L \sin\theta \quad [\text{Volt-Amperes Reactive, VAR}]$$

$$S_{3\phi} = \sqrt{P_{3\phi}^2 + Q_{3\phi}^2} = \sqrt{3}V_L I_L \quad [\text{Volt-Amperes, VA}]$$

CAUTION

In the formula $\sqrt{3}V_L I_L \cos\theta$, the angle θ is **the phase angle of the load impedance** ($\theta = \theta_v - \theta_i$ of individual phase branch), **NOT** the angle between line voltage and line current!

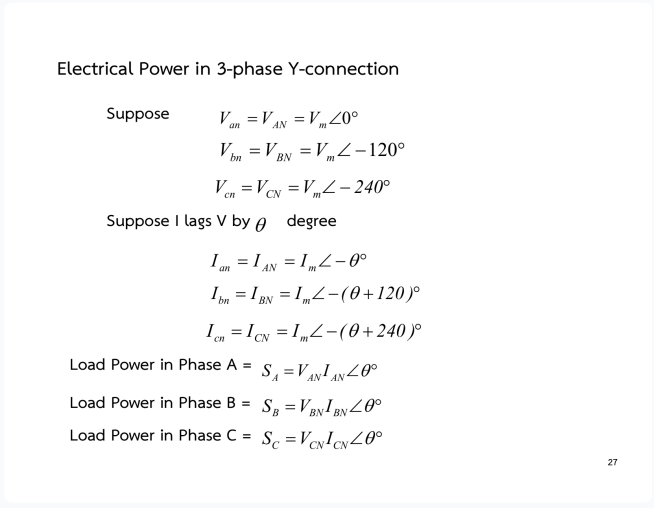


Figure 10: Three-Phase Power Formulas from Slide 27

Figure 10: Real, reactive, and apparent power formulas for balanced three-phase systems (From Slide 27).

Two-Wattmeter Method for Power Measurement

The two-wattmeter method measures the total real and reactive power of a three-phase three-wire system (balanced or unbalanced) using only two wattmeters:

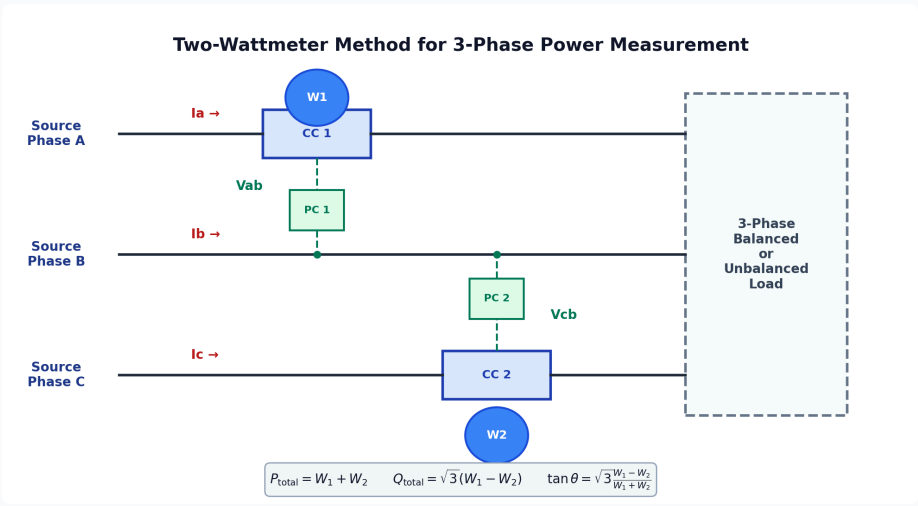


Figure 11: Schematic of the Two-Wattmeter Connection

Figure 11: Circuit diagram of the Two-Wattmeter Method with Current Coils (CC) in lines A and C, and Potential Coils (PC) connected across common line B.

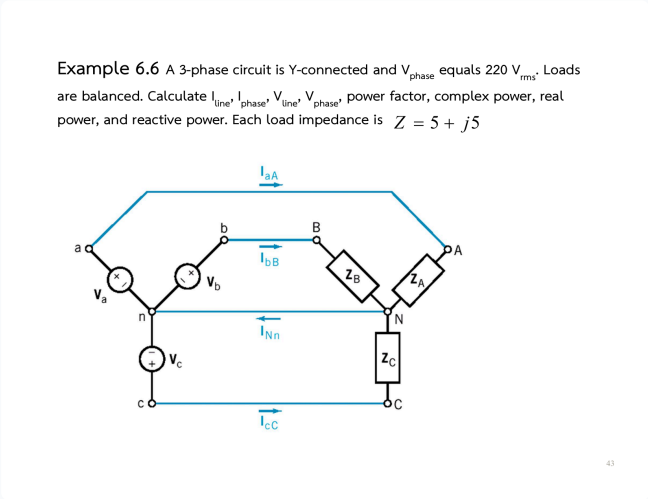


Figure 12: Two-Wattmeter Example Slide

Figure 12: Derivation and vector angles of individual wattmeter readings (From Slide 43).

1. Wattmeter Reading Equations (Balanced Load, Phase Sequence abc)

- Wattmeter 1 measures line current $\mathbf{I}_a$ and voltage $\mathbf{V}_{ab}$:

$$W_1 = V_{ab} I_a \cos(\angle \mathbf{V}_{ab} - \angle \mathbf{I}_a) = V_L I_L \cos(30^\circ - \theta)$$

- Wattmeter 2 measures line current $\mathbf{I}_c$ and voltage $\mathbf{V}_{cb}$:

$$W_2 = V_{cb} I_c \cos(\angle \mathbf{V}_{cb} - \angle \mathbf{I}_c) = V_L I_L \cos(30^\circ + \theta)$$

2. Master Formulas Derived from W_1 and W_2

1. **Total Real Power (P_{total}):**

$$P_{\text{total}} = W_1 + W_2$$

2. **Total Reactive Power (Q_{total}):**

$$Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$$

3. **Power Factor Angle (θ):**

$$\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2} \implies \text{pf} = \cos\theta$$

Load Power Factor ($\text{pf} = \cos\theta$)	Load Angle (θ)	Wattmeter 1 Reading (W_1)	Wattmeter 2 Reading (W_2)	Special Phenomenon
$\text{pf} = 1.0$ **(Purely Resistive)**	$\theta = 0^\circ$	$W_1 = W_2 > 0$	$W_1 = W_2 > 0$	Both wattmeters read identical positive values.
$\text{pf} = 0.5$ **(Lagging)**	$\theta = +60^\circ$	$W_1 > 0$	$W_2 = 0$	**Wattmeter 2 reads exactly ZERO!**
$0 \le \text{pf} < 0.5$ **(Lagging)**	$\theta > 60^\circ$	$W_1 > 0$	$W_2 < 0$ **(Negative!)**	**Wattmeter 2 deflects backwards!** Reverse CC terminals to read.
$\text{pf} = 0$ **(Purely Reactive)**	$\theta = 90^\circ$	$W_1 = -W_2$	$W_2 = -W_1$	$W_1 + W_2 = 0$ (Zero total real power).

⚙️ Power Factor Correction with Delta Capacitor Banks

Most industrial loads are inductive (induction motors, transformers) operating at lagging power factors. Installing a parallel capacitor bank supplies the required reactive power locally, unloading transmission lines:

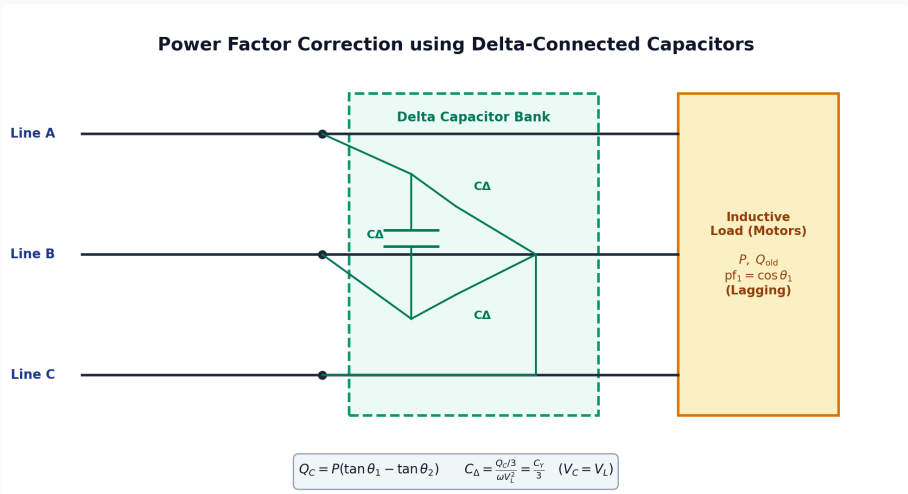


Figure 13: Power Factor Correction Schematic with Delta Capacitor Bank

Figure 13: Schematic of a three-phase inductive motor load corrected by a parallel Delta-connected capacitor bank.

1. Step-by-Step Design Formula

1. Calculate the required total reactive power compensation:

$$Q_C = P(\tan\theta_1 - \tan\theta_2)$$

2. In a **Delta-connected** capacitor bank, each capacitor sees the full line-to-line voltage ($V_C = V_L$):

$$Q_{C,\text{per phase}} = \frac{Q_C}{3} = \omega C_\Delta V_L^2 \implies C_\Delta = \frac{Q_C}{3\omega V_L^2}$$

3. If connected in **Wye (Y)**, the voltage across each capacitor is only $V_p = V_L/\sqrt{3}$:

$$C_Y = \frac{Q_C}{\omega V_L^2} = 3C_\Delta$$

TIP

Why Delta is Universally Preferred in Industry: A Delta-connected capacitor bank requires $3\times$ **smaller capacitance** than a Wye bank ($C_\Delta = \frac{1}{3}C_Y$) to provide identical reactive VARs. Smaller capacitance means smaller physical size, lighter weight, and lower equipment cost!

📝 5 Authentic Exam Problems with Full Step-by-Step Solutions

🔴 Problem 1 (Level 1: Fundamentals of Phase Sequence & Line Voltages)

Problem Statement:
A balanced three-phase Wye-connected source has a positive sequence (abc) with phase voltage $\mathbf{V}_{an} = 120\angle 10^\circ \text{ V rms}$.
Find:

1. The remaining phase voltages $\mathbf{V}_{bn}$ and $\mathbf{V}_{cn}$.
2. All three line-to-line voltages $\mathbf{V}_{ab}, \mathbf{V}_{bc}, \mathbf{V}_{ca}$.

💡 Step-by-Step Model Solution:

Step 1: Determine Phase Voltages
In positive sequence, each phase lags by 120°:

$$\mathbf{V}_{an} = 120\angle 10^\circ \text{ V}$$

$$\mathbf{V}_{bn} = 120\angle(10^\circ - 120^\circ) = 120\angle -110^\circ \text{ V}$$

$$\mathbf{V}_{cn} = 120\angle(10^\circ + 120^\circ) = 120\angle +130^\circ \text{ V}$$

Step 2: Calculate Line Voltages
Line voltage magnitude is $V_L = \sqrt{3}V_p = \sqrt{3} \times 120 \approx 207.85 \text{ V}$, leading phase voltage by 30°:

$$\mathbf{V}_{ab} = \sqrt{3} \times 120\angle(10^\circ + 30^\circ) = 207.85\angle 40^\circ \text{ V}$$

$$\mathbf{V}_{bc} = 207.85\angle(40^\circ - 120^\circ) = 207.85\angle -80^\circ \text{ V}$$

$$\mathbf{V}_{ca} = 207.85\angle(40^\circ + 120^\circ) = 207.85\angle 160^\circ \text{ V}$$

🔴 Problem 2 (Level 2: Balanced Y-Y System with Line Impedance - Homework 6 Prob. 1)

Problem Statement:
A balanced 3-phase 4-wire Y-Y system has:

- Source: $\mathbf{V}_{an} = 110\angle 0^\circ \text{ V rms}$, abc sequence.
- Line impedance: $\mathbf{Z}_{\text{line}} = 5 - j2\ \Omega$ per phase.
- Load impedance: $\mathbf{Z}_Y = 10 + j8\ \Omega$ per phase.
- Neutral wire impedance: $\mathbf{Z}_n = 0\ \Omega$.
  Find:

1. The line currents $\mathbf{I}_{aA}, \mathbf{I}_{bB}, \mathbf{I}_{cC}$.
2. The phase voltages at the load terminals $\mathbf{V}_{AN}, \mathbf{V}_{BN}, \mathbf{V}_{CN}$.

3. The total complex power $\mathbf{S}_{\text{total}}$ delivered to the load.

active power and power factor.

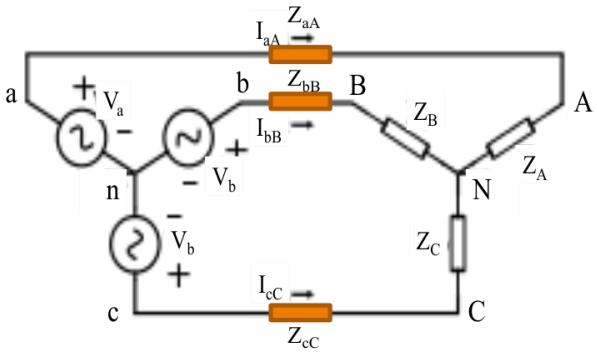


Fig. 1 Y to Y circuit with line impedance and balanced load.

Figure 14: Balanced Y-Y Circuit Diagram

Figure 14: Balanced Y-Y system with line impedance (Homework 6 Problem 1).

💡 **Step-by-Step Model Solution:**

Step 1: Single-Phase Equivalent Circuit Analysis

Total impedance per phase:

$$\mathbf{Z}_{\text{total}} = \mathbf{Z}_{\text{line}} + \mathbf{Z}_Y = (5 - j2) + (10 + j8) = 15 + j6\ \Omega$$

$$\mathbf{Z}_{\text{total}} = \sqrt{15^2 + 6^2}\angle \arctan(6/15) = \sqrt{225 + 36}\angle 21.8^\circ = 16.16\angle 21.8^\circ\ \Omega$$

Step 2: Line Current Calculation

$$\mathbf{I}_{aA} = \frac{\mathbf{V}_{an}}{\mathbf{Z}_{\text{total}}} = \frac{110\angle 0^\circ}{16.16\angle 21.8^\circ} = 6.81\angle -21.8^\circ\ \text{A}$$

$$\mathbf{I}_{bB} = 6.81\angle(-21.8^\circ - 120^\circ) = 6.81\angle -141.8^\circ\ \text{A}$$

$$\mathbf{I}_{cC} = 6.81\angle(-21.8^\circ + 120^\circ) = 6.81\angle 98.2^\circ\ \text{A}$$

$$\mathbf{I}_n = -(\mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC}) = 0\ \text{A (Balanced!)}$$

Step 3: Load Phase Voltages

$$\mathbf{V}_{AN} = \mathbf{I}_{aA}\mathbf{Z}_Y = (6.81\angle -21.8^\circ) \times (10 + j8)$$

$$10 + j8 = \sqrt{100 + 64} \angle \arctan(8/10) = 12.81 \angle 38.66^\circ \ \Omega$$

$$\mathbf{V}_{AN} = 6.81 \times 12.81 \angle (-21.8^\circ + 38.66^\circ) = 87.24 \angle 16.86^\circ \ \text{V}$$

$$\mathbf{V}_{BN} = 87.24 \angle (16.86^\circ - 120^\circ) = 87.24 \angle -103.14^\circ \ \text{V}$$

$$\mathbf{V}_{CN} = 87.24 \angle (16.86^\circ + 120^\circ) = 87.24 \angle 136.86^\circ \ \text{V}$$

Step 4: Total Complex Power Delivered to the Load

$$\mathbf{S}_{3\phi} = 3 \times |\mathbf{I}_{aA}|^2 \times \mathbf{Z}_Y = 3 \times (6.81)^2 \times (10 + j8) = 3 \times 46.38 \times (10 + j8)$$

$$\mathbf{S}_{3\phi} = 1,391.4 + j1,113.1 \ \text{VA} = 1.39 \ \text{kW} + j1.11 \ \text{kVAR}$$

🔴 Problem 3 (Level 3: Balanced Y-Δ System with Line Impedance)

Problem Statement:
A balanced three-phase system has:

- Source: Wye-connected, line-to-line voltage $V_L = 208 \ \text{V rms}$, abc sequence, taking $\mathbf{V}_{ab}$ as reference ($208 \angle 0^\circ \ \text{V}$).
- Line impedance: $\mathbf{Z}_{\text{line}} = 1 + j1 \ \Omega$.
- Load: Delta-connected with branch impedance $\mathbf{Z}_\Delta = 12 + j9 \ \Omega$.
  Find:

1. The line currents $\mathbf{I}_a, \mathbf{I}_b, \mathbf{I}_c$.
2. The phase currents circulating inside the Delta load $\mathbf{I}_{AB}, \mathbf{I}_{BC}, \mathbf{I}_{CA}$.

💡 **Step-by-Step Model Solution:**

Step 1: Transform Delta Load to Equivalent Wye Load

$$\mathbf{Z}_Y = \frac{\mathbf{Z}_\Delta}{3} = \frac{12 + j9}{3} = 4 + j3 \ \Omega$$

Step 2: Determine Source Phase Voltage Since $\mathbf{V}_{ab} = 208 \angle 0^\circ \ \text{V}$ and $\mathbf{V}_{ab} = \sqrt{3} \mathbf{V}_{an} \angle +30^\circ$:

$$\mathbf{V}_{an} = \frac{208}{\sqrt{3}} \angle (0^\circ - 30^\circ) = 120.09 \angle -30^\circ \ \text{V}$$

Step 3: Total Impedance and Line Current

$$\mathbf{Z}_{\text{total}} = \mathbf{Z}_{\text{line}} + \mathbf{Z}_Y = (1 + j1) + (4 + j3) = 5 + j4 = 6.403\angle 38.66^\circ\ \Omega$$

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_{\text{total}}} = \frac{120.09\angle -30^\circ}{6.403\angle 38.66^\circ} = 18.755\angle -68.66^\circ\ \text{A}$$

$$\mathbf{I}_b = 18.755\angle(-68.66^\circ - 120^\circ) = 18.755\angle -188.66^\circ\ \text{A} = 18.755\angle 171.34^\circ\ \text{A}$$

$$\mathbf{I}_c = 18.755\angle(-68.66^\circ + 120^\circ) = 18.755\angle 51.34^\circ\ \text{A}$$

Step 4: Calculate Delta Phase Currents Phase current magnitude is $I_p = I_L/\sqrt{3} = 18.755/\sqrt{3} = 10.828\ \text{A}$, leading line current by 30°:

$$\mathbf{I}_{AB} = \frac{18.755}{\sqrt{3}}\angle(-68.66^\circ + 30^\circ) = 10.83\angle -38.66^\circ\ \text{A}$$

$$\mathbf{I}_{BC} = 10.83\angle(-38.66^\circ - 120^\circ) = 10.83\angle -158.66^\circ\ \text{A}$$

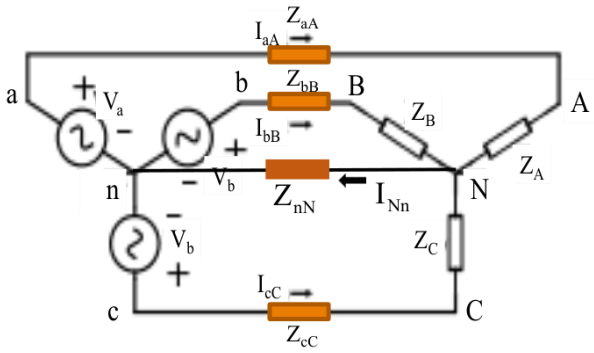
$$\mathbf{I}_{CA} = 10.83\angle(-38.66^\circ + 120^\circ) = 10.83\angle 81.34^\circ\ \text{A}$$

🔴 Problem 4 (Level 4: Unbalanced Y-Y System & Neutral Shift - Homework 6 Prob. 2)

Problem Statement:
An unbalanced 3-phase 3-wire Y-Y system (NO neutral wire, $\mathbf{Z}_n = \infty$) has:

- Source: $\mathbf{V}_{an} = 100\angle 0^\circ\ \text{V}$, $\mathbf{V}_{bn} = 100\angle -120^\circ\ \text{V}$, $\mathbf{V}_{cn} = 100\angle 120^\circ\ \text{V}$.
- Line impedance: $\mathbf{Z}_{\text{line}} = 0\ \Omega$.
- Unbalanced load: $\mathbf{Z}_A = 10\ \Omega$, $\mathbf{Z}_B = j10\ \Omega$, $\mathbf{Z}_C = -j10\ \Omega$.
  Find:

1. The Neutral Shift Voltage ($\mathbf{V}_{Nn} = \mathbf{V}_N - \mathbf{V}_n$).
2. The load phase voltages $\mathbf{V}_{AN}, \mathbf{V}_{BN}, \mathbf{V}_{CN}$.
3. The line currents $\mathbf{I}_{aA}, \mathbf{I}_{bB}, \mathbf{I}_{cC}$.



ig. 2 Y to Y circuit with line impedance and unbalanced load.

ed from: https://en.wikipedia.org/wiki/Three-phase_electric_power

Figure 15: Unbalanced Y-Y Circuit Diagram

Figure 15: Unbalanced Y-Y load with disconnected neutral wire (Homework 6 Problem 2).

💡 **Step-by-Step Model Solution:**

Step 1: Calculate Load Admittances

$$\mathbf{Y}_A = \frac{1}{10} = 0.1 \text{ S}$$

$$\mathbf{Y}_B = \frac{1}{j10} = -j0.1 \text{ S}$$

$$\mathbf{Y}_C = \frac{1}{-j10} = +j0.1 \text{ S}$$

Sum of admittances:

$$\sum \mathbf{Y} = \mathbf{Y}_A + \mathbf{Y}_B + \mathbf{Y}_C = 0.1 - j0.1 + j0.1 = 0.1 \text{ S}$$

Step 2: Calculate Neutral Shift Voltage using Millman's Theorem / KCL

$$\mathbf{V}_{Nn} = \frac{\mathbf{V}_{an}\mathbf{Y}_A + \mathbf{V}_{bn}\mathbf{Y}_B + \mathbf{V}_{cn}\mathbf{Y}_C}{\sum \mathbf{Y}}$$

Numerator:

$$\mathbf{V}_{an}\mathbf{Y}_A = (100\angle 0^\circ)(0.1) = 10$$

$$\mathbf{V}_{bn}\mathbf{Y}_B = (100\angle -120^\circ)(-j0.1) = (100\angle -120^\circ)(0.1\angle -90^\circ) = 10\angle -210^\circ = 10\angle 150^\circ = -8.66 + j5$$

$$\mathbf{V}_{cn}\mathbf{Y}_C = (100\angle 120^\circ)(+j0.1) = (100\angle 120^\circ)(0.1\angle 90^\circ) = 10\angle 210^\circ = -8.66 - j5$$

Summing the terms:

$$\text{Numerator} = 10 + (-8.66 + j5) + (-8.66 - j5) = 10 - 17.32 = -7.32 \text{ V}$$

$$\mathbf{V}_{Nn} = \frac{-7.32}{0.1} = -73.2\angle 0^\circ = 73.2\angle 180^\circ \text{ V}$$

Step 3: Calculate Load Phase Voltages

$$\mathbf{V}_{AN} = \mathbf{V}_{an} - \mathbf{V}_{Nn} = 100 - (-73.2) = 173.2\angle 0^\circ \text{ V}$$

$$\mathbf{V}_{BN} = \mathbf{V}_{bn} - \mathbf{V}_{Nn} = (-50 - j86.6) - (-73.2) = 23.2 - j86.6 = 89.65\angle -75^\circ \text{ V}$$

$$\mathbf{V}_{CN} = \mathbf{V}_{cn} - \mathbf{V}_{Nn} = (-50 + j86.6) - (-73.2) = 23.2 + j86.6 = 89.65\angle 75^\circ \text{ V}$$

Step 4: Calculate Line Currents

$$\mathbf{I}_{aA} = \frac{\mathbf{V}_{AN}}{\mathbf{Z}_A} = \frac{173.2\angle 0^\circ}{10} = 17.32\angle 0^\circ \text{ A}$$

$$\mathbf{I}_{bB} = \frac{\mathbf{V}_{BN}}{\mathbf{Z}_B} = \frac{89.65\angle -75^\circ}{10\angle 90^\circ} = 8.965\angle -165^\circ \text{ A}$$

$$\mathbf{I}_{cC} = \frac{\mathbf{V}_{CN}}{\mathbf{Z}_C} = \frac{89.65\angle 75^\circ}{10\angle -90^\circ} = 8.965\angle 165^\circ \text{ A}$$

(Verification: $\mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC} = 17.32 + 2(-8.66) = 0$ A. Perfect KCL match!)

🔴 Problem 5 (Level 5: Two-Wattmeter Method & Power Factor Correction Exam Problem)

Problem Statement:
A three-phase, 400 V rms, 50 Hz supply feeds a balanced inductive load. Two wattmeters connected to measure total input power read:

$$W_1 = 12.5 \text{ kW} \quad \text{and} \quad W_2 = 4.5 \text{ kW}$$

1. Find the total real power P_{total}, reactive power Q_{total}, apparent power S_{total}, and the power factor.
2. Determine the line current I_L.

3. Calculate the required capacitance per phase (C_Δ) of a Delta-connected capacitor bank to improve the overall power factor to 0.95 lagging.

💡 **Step-by-Step Model Solution:**

Step 1: Calculate Total Powers from Wattmeter Readings

$$P_{\text{total}} = W_1 + W_2 = 12.5 + 4.5 = 17.0 \text{ kW}$$

$$Q_{\text{total}} = \sqrt{3}(W_1 - W_2) = \sqrt{3}(12.5 - 4.5) = \sqrt{3} \times 8.0 \approx 13.856 \text{ kVAR}$$

$$S_{\text{total}} = \sqrt{P^2 + Q^2} = \sqrt{17.0^2 + 13.856^2} = \sqrt{289 + 192} = \sqrt{481} \approx 21.93 \text{ kVA}$$

$$\text{pf}_1 = \cos\theta_1 = \frac{P_{\text{total}}}{S_{\text{total}}} = \frac{17.0}{21.93} \approx 0.775 \text{ lagging}$$

$$\theta_1 = \arccos(0.775) \approx 39.18^\circ$$

Step 2: Line Current Before Compensation

$$I_L = \frac{P_{\text{total}}}{\sqrt{3} V_L \cos\theta_1} = \frac{17,000}{\sqrt{3} \times 400 \times 0.775} = \frac{17,000}{536.94} \approx 31.66 \text{ A}$$

Step 3: Capacitor Bank Sizing for $\text{pf}_2 = 0.95$ lagging

$$\theta_2 = \arccos(0.95) \approx 18.19^\circ$$

$$\tan\theta_1 = \tan(39.18^\circ) = 0.815$$

$$\tan\theta_2 = \tan(18.19^\circ) = 0.329$$

Required reactive power compensation:

$$Q_C = P_{\text{total}}(\tan\theta_1 - \tan\theta_2) = 17.0 \times (0.815 - 0.329) = 17.0 \times 0.486 = 8.262 \text{ kVAR}$$

For a **Delta-connected** capacitor bank ($V_C = V_L = 400$ V):

$$C_\Delta = \frac{Q_C/3}{\omega V_L^2} = \frac{8,262/3}{2\pi \times 50 \times 400^2} = \frac{2,754}{2\pi \times 50 \times 160,000} = \frac{2,754}{50,265,482} \approx 5.48 \times 10^{-5} \text{ F} = 54.8 \ \mu\text{F}$$

🎯 Final Exam Key Formulas & Cheat Sheet

1. **Wye (Y) Relationships**:
   - $V_L = \sqrt{3} V_p \angle +30^\circ$
   - $I_L = I_p$
2. **Delta (Δ) Relationships**:
   - $V_L = V_p$
   - $I_L = \sqrt{3} I_p \angle -30^\circ$
3. **$\Delta \leftrightarrow Y$ Impedance Transformation**:
   - $Z_Y = \frac{Z_\Delta}{3} \iff Z_\Delta = 3Z_Y$
4. **Three-Phase Power**:
   - $P = \sqrt{3} V_L I_L \cos\theta = 3 V_p I_p \cos\theta$
   - $Q = \sqrt{3} V_L I_L \sin\theta = 3 V_p I_p \sin\theta$
5. **Two-Wattmeter Method**:
   - $P_{\text{total}} = W_1 + W_2$
   - $Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$
   - $\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$
6. **Capacitor Sizing**:
   - $Q_C = P(\tan\theta_1 - \tan\theta_2)$
   - $C_\Delta = \frac{Q_C/3}{\omega V_L^2} = \frac{C_Y}{3}$